Lakshya JEE 2.0 2025

Maths

Basic Mathematics

DPP: 2

Q1 Find the principal value of the trigonometric function of $\cos x=\frac{\sqrt{3}}{2}$.

(A) $\frac{\pi}{4}, \frac{5\pi}{4}$

(B) $\frac{\pi}{4}, \frac{3\pi}{4}$

(C) $\frac{\pi}{3}, \frac{2\pi}{3}$

(D) $\frac{\pi}{6}, \frac{11\pi}{6}$

Q2 Find the principal solutions of the equation $\tan x=-\frac{1}{\sqrt{3}}$

(A) $\frac{5\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{\pi}{6}, \frac{11\pi}{6}$

(C) $\frac{7\pi}{6}$

(D) None of these

Q3 $\cot\theta+\tan\theta=2\operatorname{cosec}\theta$, then find principal value of θ.

(A) $\frac{\pi}{3}$ and $\frac{5\pi}{3}$

(B) 0 and $\frac{2\pi}{3}$

(C) $\frac{\pi}{3}$ and $\frac{\pi}{6}$

(D) $\frac{\pi}{6}$ and $\frac{2\pi}{3}$

Q4 Find the general solution of the following equation: $2\sin\theta+1=0$

(A) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{\pi}{6}$

(B) $\mathrm{n}\pi+(-1)^{\mathrm{n}+1}\frac{\pi}{6}$

(C) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{5\pi}{6}$

(D) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{\pi}{3}$

Q5 Find the general solution of the following equation: $\cos 3\theta=-\frac{1}{2}$

(A) $2\mathrm{n}\pi\pm\frac{2\pi}{3}$

(B) $2\mathrm{n}\pi\pm\frac{\pi}{3}$

(C) $\frac{2n\pi}{3}\pm\frac{2\pi}{9}$

(D) $\frac{2n\pi}{3}\pm\frac{4\pi}{3}$

Q6 Find the general solution of the following equation : $\sqrt{3}\sec\theta=2$

(A) $\theta=2\mathrm{n}\pi\pm\frac{\pi}{3}$

(B) $\theta=2\mathrm{n}\pi\pm\frac{\pi}{6}$

(C) $\theta=2\mathrm{n}\pi\pm\frac{5\pi}{6}$

(D) $\theta=2\mathrm{n}\pi\pm\frac{7\pi}{6}$

Q7 Write the general solution of the trigonometric equation $4\tan^2\theta=3\sec^2\theta$

(A) $n\pi\pm\frac{\pi}{6}$

(B) $\mathrm{n}\pi\pm\frac{\pi}{3}$

(C) $\mathrm{n}\pi\pm\frac{5\pi}{6}$

(D) $\mathrm{n}\pi\pm\frac{2\pi}{3}$

Q8 If $\frac{1}{6}\sin\theta, \cos\theta$ and $\tan\theta$ are in $\mathrm{G.P.}$ then the general solution for θ is

(A) $2\mathrm{n}\pi\pm\frac{\pi}{3}$

(B) $2n\pi\pm\frac{\pi}{6}$

(C) $\mathrm{n}\pi\pm\frac{\pi}{3}$

(D) None of these

Q9 Solution of the equation is $\sin x+\cos x=\sqrt{2}$

(A) $2\mathrm{n}\pi-\frac{\pi}{4}$

(B) $\mathrm{n}\pi+\frac{\pi}{4}$

(C) $2\mathrm{n}\pi+\frac{\pi}{4}$

(D) $\mathrm{n}\pi-\frac{\pi}{4}$

Q10 If $\sqrt{3}\cos\theta+\sin\theta=\sqrt{2}$, then the most general value of θ is

(A) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{\pi}{4}$

(B) $(-1)^{\mathrm{n}}\frac{\pi}{4}-\frac{\pi}{3}$

(C) $\mathrm{n}\pi+\frac{\pi}{4}-\frac{\pi}{3}$

(D) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$

Q11 Solve the following trigonometric equation $\tan\left(\frac{2}{3}\theta\right) = \sqrt{3}$

(A) $\theta = n\pi + \frac{\pi}{3}$

(B) $\theta = \frac{3n\pi}{2} + \frac{\pi}{2}$

(C) $\theta = n\pi + \frac{\pi}{2}$

(D) $\theta = \frac{3n\pi}{2} - \frac{\pi}{2}$

Q12 If $4\sin^2\theta = 1$, then the value of θ are-

(A) $2n\pi \pm \frac{\pi}{3}, n \in Z$

(B) $n\pi \pm \frac{\pi}{3}, n \in Z$

(C) $n\pi \pm \frac{\pi}{6}, n \in Z$

(D) $2n\pi \pm \frac{\pi}{6}, n \in Z$

Q13 The number of solution of the equation $\tan x + \sec x = 2\cos x$ lying in the interval $(0, 2\pi)$ is.

(A) 0 (B) 1

(C) 2 (D) 3

Q14 The number of solutions of the equation $3\sin^2 x - 7\sin x + 2 = 0$ in the interval $[0, 5\pi]$,is

(A) 0 (B) 5

(C) 6 (D) 10

Q15 The number of solution in $[0, 2\pi]$ of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, is

(A) 2 (B) 4

(C) 6 (D) 8

Q16 If $\sin 6\theta + \sin 4\theta + \sin 2\theta = 0$, Then the general value of θ is -

(A) $\frac{n\pi}{4}, n\pi \pm \pi/3$

(B) $\frac{n\pi}{4}, 2n\pi \pm \pi/3$

(C) $\frac{n\pi}{4}, n\pi \pm \pi/6$

(D) $\frac{n\pi}{4}, 2n\pi \pm \pi/6$

Q17 Solve the equation & find the value of x $\tan\left(\frac{\pi}{4} - x\right) + \tan\left(\frac{\pi}{4} + x\right) = 4$

(A) $x = n\pi \pm \frac{\pi}{3}$

(B) $x = n\pi \pm \frac{\pi}{6}$

(C) $x = n\pi + \frac{\pi}{3}$

(D) $x = n\pi + \frac{\pi}{6}$

Answer Key

Q1	(D)
Q2	(A)
Q3	(A)
Q4	(B)
Q5	(C)
Q6	(B)
Q7	(B)
Q8	(A)
Q9	(C)
Q10	(D)
Q11	(B)
Q12	(C)
Q13	(C)
Q14	(C)
Q15	(D)
Q16	(A)
Q17	(B)